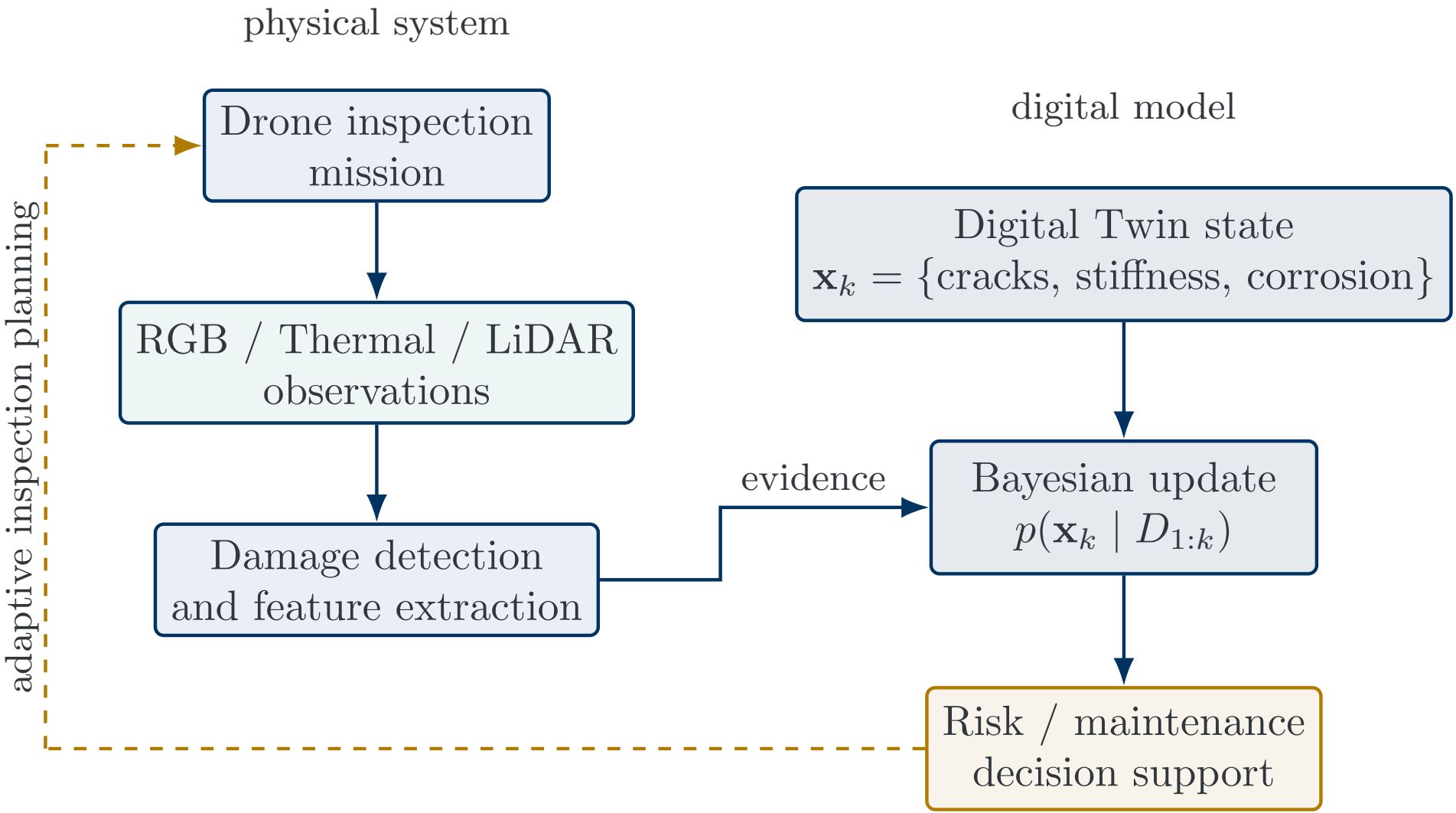

physical system
digital model
Drone inspection mission
RGB / Thermal / LiDAR observations
Damage detection and feature extraction
Digital Twin state
$\mathbf{x}_k = \{\text{cracks, stiffness, corrosion}\}$
evidence
Bayesian update
$p(\mathbf{x}_k \mid D_{1:k})$
Risk / maintenance decision support
adaptive inspection planning